

Robert, 58 · sudden crushing substernal chest pain · diaphoretic, nauseated, radiation to left jaw and arm · BP 150/95, HR 105 · EKG: ST elevation in leads II, III, aVF · **inferior STEMI**

BIO 004 · WEEK 3 · CASE DEEP DIVE

The Heart

A 58-year-old grabs his chest at his desk. The infarcted territory tells you which artery clogged.

PATIENT CASE

Robert is 58, hypertensive, hyperlipidemic, sedentary, 30-pack-year smoker. He is at his desk at 10 AM when he develops **sudden crushing substernal chest pain**, "like an elephant sitting on my chest," radiating to his **left jaw and down the inside of the left arm**. He is **diaphoretic**, nauseated, and short of breath.

In the ED his BP is 150/95, HR 105, sat 96% on room air. EKG shows **ST-segment elevation in leads II, III, and aVF**. Troponin is sent and the cath lab is activated. Robert is having an **inferior wall ST-elevation myocardial infarction (STEMI)**, almost certainly from occlusion of his **right coronary artery**.

To understand what is dying in his heart, why it shows up on three specific EKG leads, and why his pulse is changing, you have to know how the heart is built: its four chambers, four valves, two pumps in one organ, three layers in its wall, and the two coronary arteries that feed it.

GUIDING QUESTION

How is the heart organized into chambers and valves, what is the pathway of blood through the two circuits, and how does the coronary artery distribution map onto what gets infarcted when one of those arteries occludes?

PREWORK

Companion to your Week 3 Heart prework page. Terms, video, and spaced recall for the chambers, valves, layers, pericardium, great vessels, and coronary circulation live there.

1 POSITION

A fist-sized pump in the middle mediastinum

The heart sits in the **middle mediastinum**, between the lungs, slightly to the left of midline, resting on the diaphragm. Roughly the size of a closed fist. The pointed inferior tip (**apex**) points down and to the left; the broad superior surface (**base**) is where the great vessels enter and leave.

The apex sits at roughly the 5th intercostal space, midclavicular line — where you palpate the point of maximal impulse (PMI) and listen for the mitral valve.

2 LAYERS & PERICARDIUM

A double-walled sac and a three-layered wall

The heart is enclosed in the **pericardium**, a double-layered serous sac:



Fibrous pericardium

Tough outer layer. Anchors the heart in the mediastinum.



Parietal layer of serous pericardium

Lines the fibrous pericardium.



Visceral layer (epicardium)

Covers the heart surface itself. Same as the outer layer of the heart wall.



Pericardial cavity

Thin space between parietal and visceral layers, containing serous fluid. Reduces friction during beating.

The heart wall has three layers from outside in: **epicardium** (= visceral pericardium), **myocardium** (the thick muscular layer that contracts; what dies in an MI), and **endocardium** (smooth inner lining continuous with the lining of vessels).

Two pumps in one organ

The heart has four chambers: two thin-walled **atria** on top that receive blood, and two thick-walled **ventricles** below that pump it out. The *right side* pumps deoxygenated blood to the lungs (pulmonary circuit). The *left side* pumps oxygenated blood to the body (systemic circuit). The two sides are separated by the *interatrial septum* and *interventricular septum*.

Four valves keep blood moving in one direction:

VALVE	TYPE	LOCATION	SOUND
Tricuspid (right AV)	3 cusps	RA → RV	contributes to S1
Pulmonary (semilunar)	3 cusps	RV → pulmonary trunk	contributes to S2
Mitral (bicuspid, left AV)	2 cusps	LA → LV	contributes to S1
Aortic (semilunar)	3 cusps	LV → aorta	contributes to S2

AV valves are held closed against ventricular pressure by **chordae tendineae** attached to **papillary muscles** in the ventricular walls. Inferior MIs (Robert's) commonly damage the papillary muscles of the right ventricle and posteromedial papillary muscle of the left ventricle, which can cause acute mitral regurgitation.

Trace it from vena cava to aorta

Memorize this loop:

1. Deoxygenated blood returns from body via **superior and inferior vena cava** → **right atrium**
2. Through **tricuspid valve** → **right ventricle**
3. Through **pulmonary valve** → **pulmonary trunk** → pulmonary arteries → lungs
4. Oxygenated blood returns via **4 pulmonary veins** → **left atrium**
5. Through **mitral valve** → **left ventricle**
6. Through **aortic valve** → **aorta** → body

The left ventricle has a much thicker wall than the right because it pumps against systemic pressure (~120 mmHg) versus the pulmonary circuit (~25 mmHg). When the left ventricle fails, blood backs up into the lungs (pulmonary edema). When the right ventricle fails, blood backs up systemically (peripheral edema, JVD).

Two arteries that feed the heart itself

The myocardium has its own dedicated blood supply. Two coronary arteries arise from the **aortic root** just above the aortic valve:

● Right coronary artery (RCA)

Runs in the right AV groove. In most people supplies: right atrium, right ventricle, the SA node (~60%), the AV node (~90%), the inferior wall of the LV, and the posterior interventricular septum.
Robert's blocked artery.

● Left main coronary artery

Divides quickly into LAD and LCX.

● Left anterior descending (LAD)

Anterior interventricular groove. Supplies: anterior wall of LV, anterior 2/3 of interventricular septum, apex. "Widow-maker" if proximally occluded.

● Left circumflex (LCX)

Left AV groove. Supplies: lateral wall of LV, left atrium.

Coronary blood return: veins drain into the **coronary sinus**, which empties into the right atrium.

Striated, branched, electrically coupled

The myocardium is made of **cardiac muscle** — striated like skeletal muscle (organized into sarcomeres) but with three critical differences: cells are *short, branched*, contain *one or two central nuclei* (not many peripheral ones like skeletal muscle), and are connected end-to-end by specialized junctions called **intercalated discs**.

Intercalated discs contain *desmosomes* (mechanical anchors that keep cells from pulling apart during contraction) and *gap junctions* (electrical connections that let action potentials propagate from cell to cell). The gap junctions are why the heart contracts as a coordinated unit (a syncytium) without needing a nerve to fire each cell individually.



DRAW ZONE 1 -- Sketch the heart in a frontal section. Label the 4 chambers, 4 valves, great vessels, and pericardial layers. Trace the pathway of blood with arrows.

Use blue arrows for deoxygenated, red for oxygenated. The crossing pattern at the ventricles is critical.



DRAW ZONE 2 -- Sketch the anterior surface of the heart with coronary arteries. Label RCA, LAD, LCX. Then shade the region of myocardium served by the RCA and circle where Robert's infarct is killing tissue.

The RCA territory is the inferior wall and posterior septum. EKG leads II, III, aVF look at exactly that.

RETURNING TO ROBERT

EKG leads, infarcted territory, the artery, the consequences

An inferior STEMI is anatomically beautiful: the EKG leads II, III, and aVF look at the inferior wall of the left ventricle, which is fed by the RCA. ST elevation in those three leads means the inferior wall is acutely ischemic, which means the RCA is occluded.

"Crushing chest pain radiating to left jaw and arm"

Visceral pain from the heart shares spinal cord levels with somatic afferents from C3-T5. The brain refers the pain to dermatomes overlapping with chest, left arm, jaw.

"ST elevation in II, III, aVF"

These three leads view the **inferior wall** of the LV. ST elevation in those leads = acute transmural ischemia of the inferior wall = RCA occlusion.

"HR 105 (mild tachycardia)"

Compensatory response to reduced cardiac output and sympathetic surge from pain. But watch closely — RCA also feeds the SA and AV nodes, so Robert is at high risk for bradycardia, AV block, or junctional rhythm as the nodes infarct.

"Diaphoretic and nauseated"

Massive sympathetic and parasympathetic activation. Vagal stimulation from inferior wall ischemia commonly produces nausea, which is why inferior MIs can be mistaken for GI complaints.

"Why time matters"

Myocardium starts dying within 20-30 minutes of occlusion. "Time is myocardium." Door-to-balloon time for PCI is the standard quality measure — < 90 minutes.

"Risk of papillary muscle rupture"

The posteromedial papillary muscle of the mitral valve has a single blood supply (from the RCA in most people). RCA occlusion can lead to papillary muscle rupture in 2-7 days, producing acute mitral regurgitation and flash pulmonary edema.

Chamber, valve, coronary, territory. Robert's EKG is a map of his heart, and the map points at the RCA. When you go back to the prework, the coronary distribution diagram will land differently — it's what the cath team is staring at right now.